

The insurance problem

See Rothschild and Stiglitz (1976).

Outline of the game

- Nature determines whether player 1 is low or high risk.
- Player 1 privately observes propensity for accident.
- Player 2, the insurance market or insurance firm, offers insurance contracts.
- Player 1 chooses a contract or remain uninsured.

Player 1: agent

- The agent has an accident with probability η ; η is private information. There are two states, accident or no accident.
- The agent may be low or high risk, identified by

$$\eta \in \{\eta_\ell, \eta_h\}, \quad 0 < \eta_\ell < \eta_h < 1.$$

- The probability that an agent is type η_h is $\lambda \in (0, 1)$.
- Agents are strictly risk averse with utility U .
- In case of accident, the agent incurs a loss ℓ . Thus, wealth is a pair $(w_a, w_n) = (w - \ell, w)$ indicating wealth in each state.

Player 2: firms or market

- An *insurance contract* is a pair (r, c) ; r is the premium and c is compensation net of premium if an accident takes place.
- Profit of a contract (r, c) sold to an agent of type η is

$$\pi(r, c, \eta) = (1 - \eta)r + \eta c.$$

- Zero-profit condition: $r = \frac{-\eta}{1-\eta} c$

Player 2; modeling alternatives

1. Competitive insurance market characterized by zero-profit condition; short run and long run.
2. At least two risk-neutral firms offering contracts under Bertrand competition.

Agent's problem

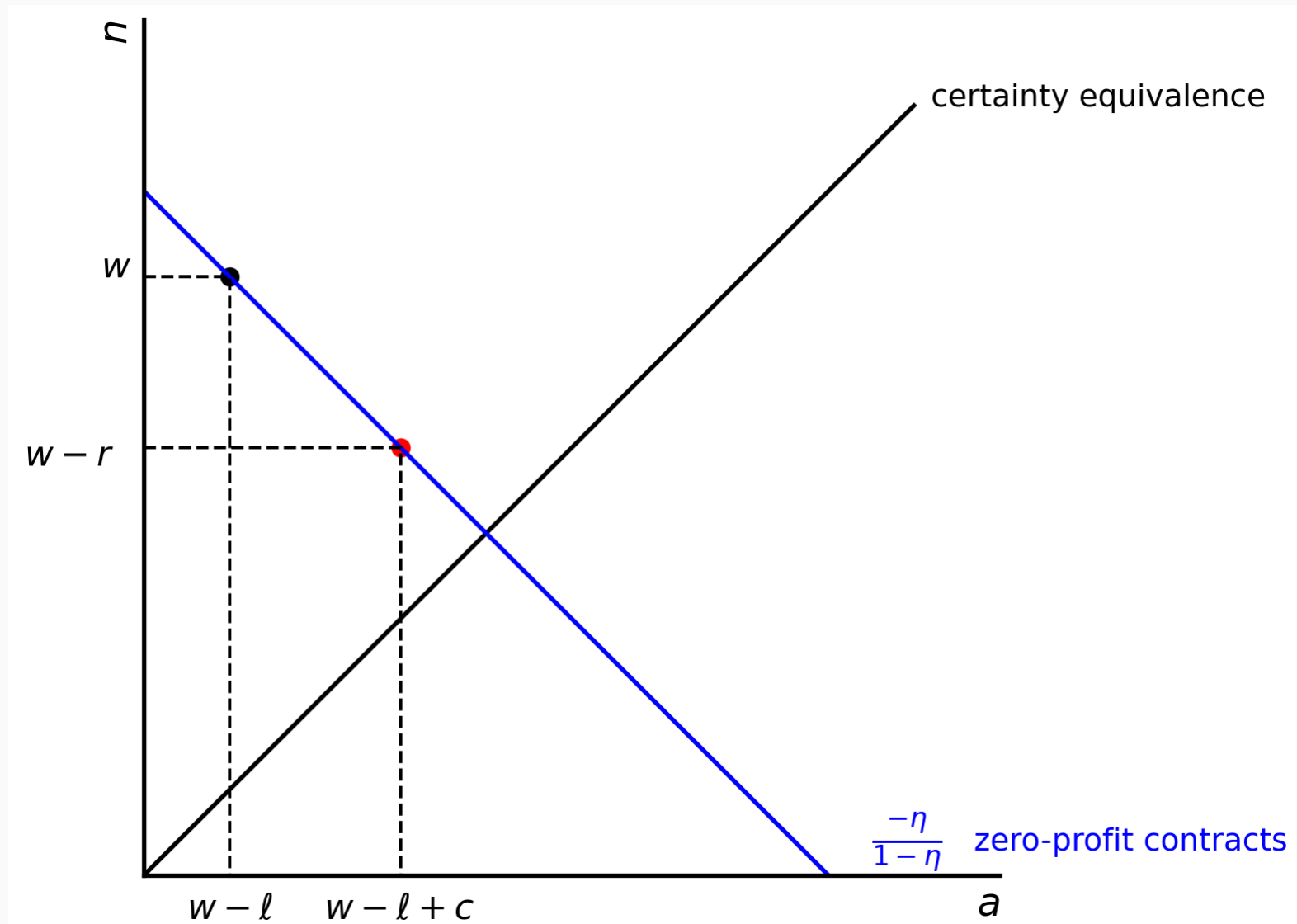
- Given (r, c) , the agent's wealth is $(w_a, w_n) = (w - \ell + c, w - r)$.
- The agent's utility is

$$V(w, r, c) = (1 - \eta)U(w - r) + \eta U(w - \ell + c).$$

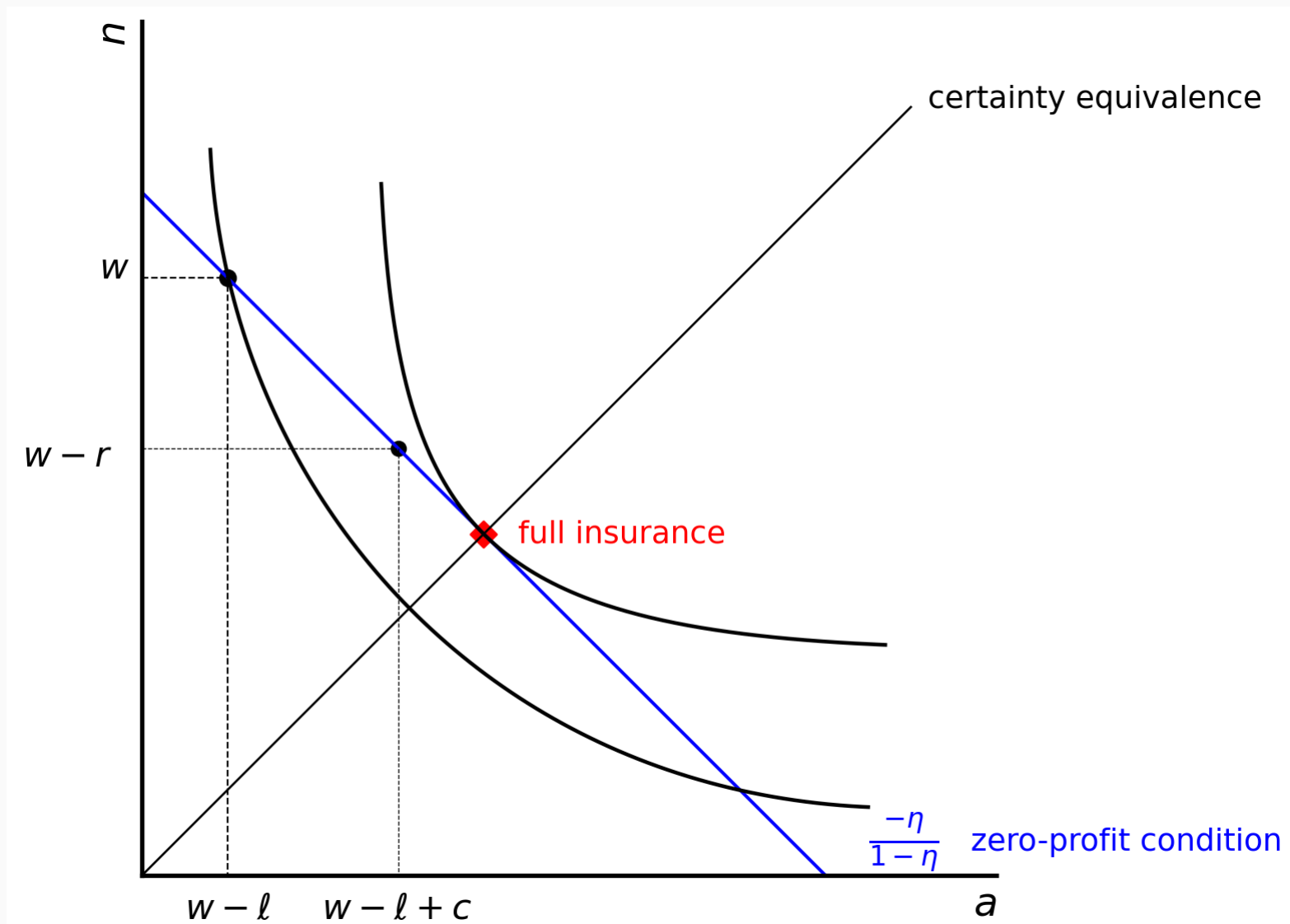
- The marginal rate of substitution

$$\begin{aligned}\bar{V} &= (1 - \eta)U(w_n) + \eta U(w_a) \\ d\bar{V} &= (1 - \eta)U'(w_n) dw_n + \eta U'(w_a) dw_a \\ \left. \frac{dw_n}{dw_a} \right|_{\bar{V}} &= -\frac{\eta}{(1 - \eta)} \frac{U'(w_a)}{U'(w_n)}\end{aligned}$$

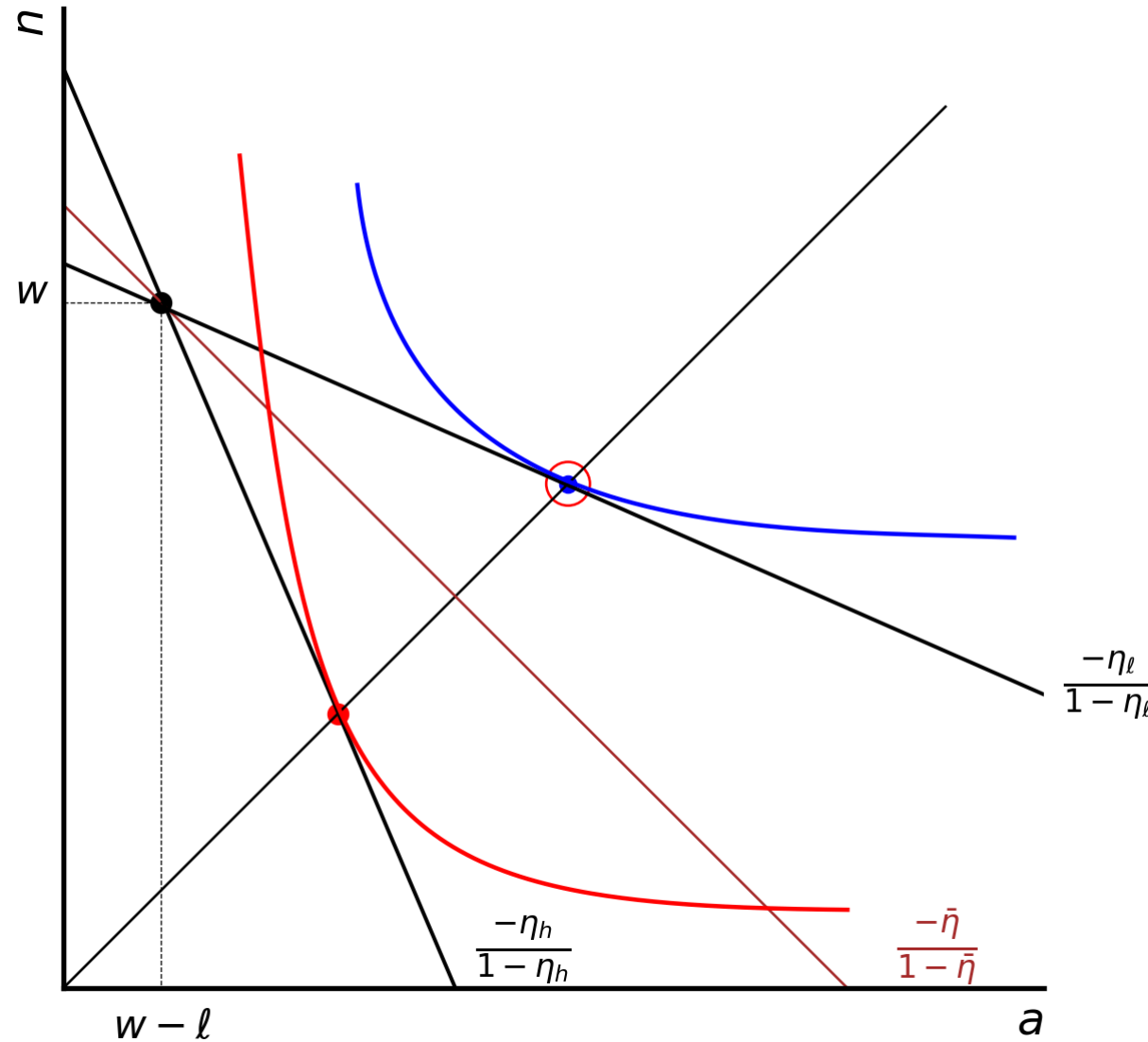
Zero profit contracts



Preferences for insurance contracts

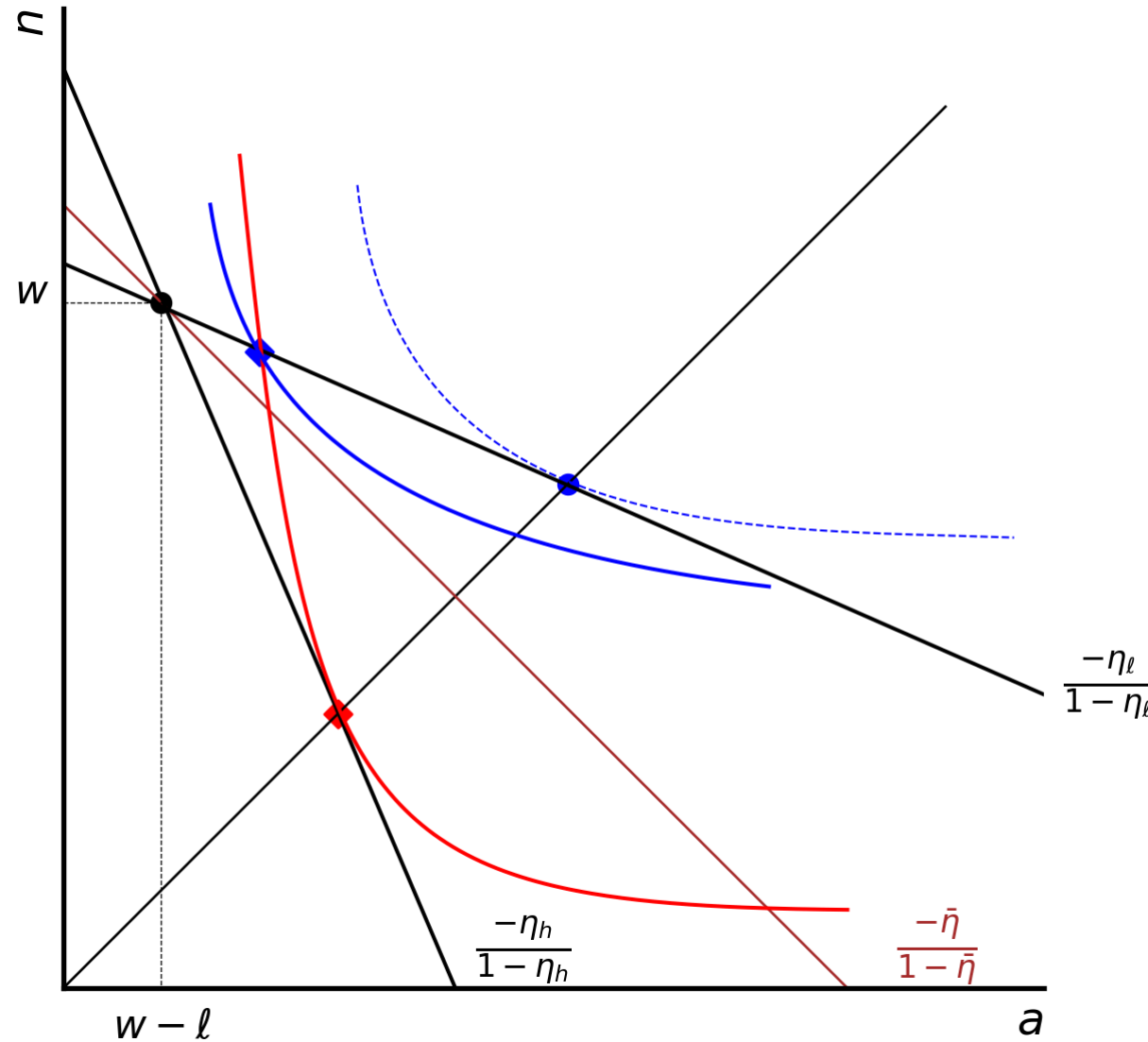


Separating equilibrium with full-insurance?



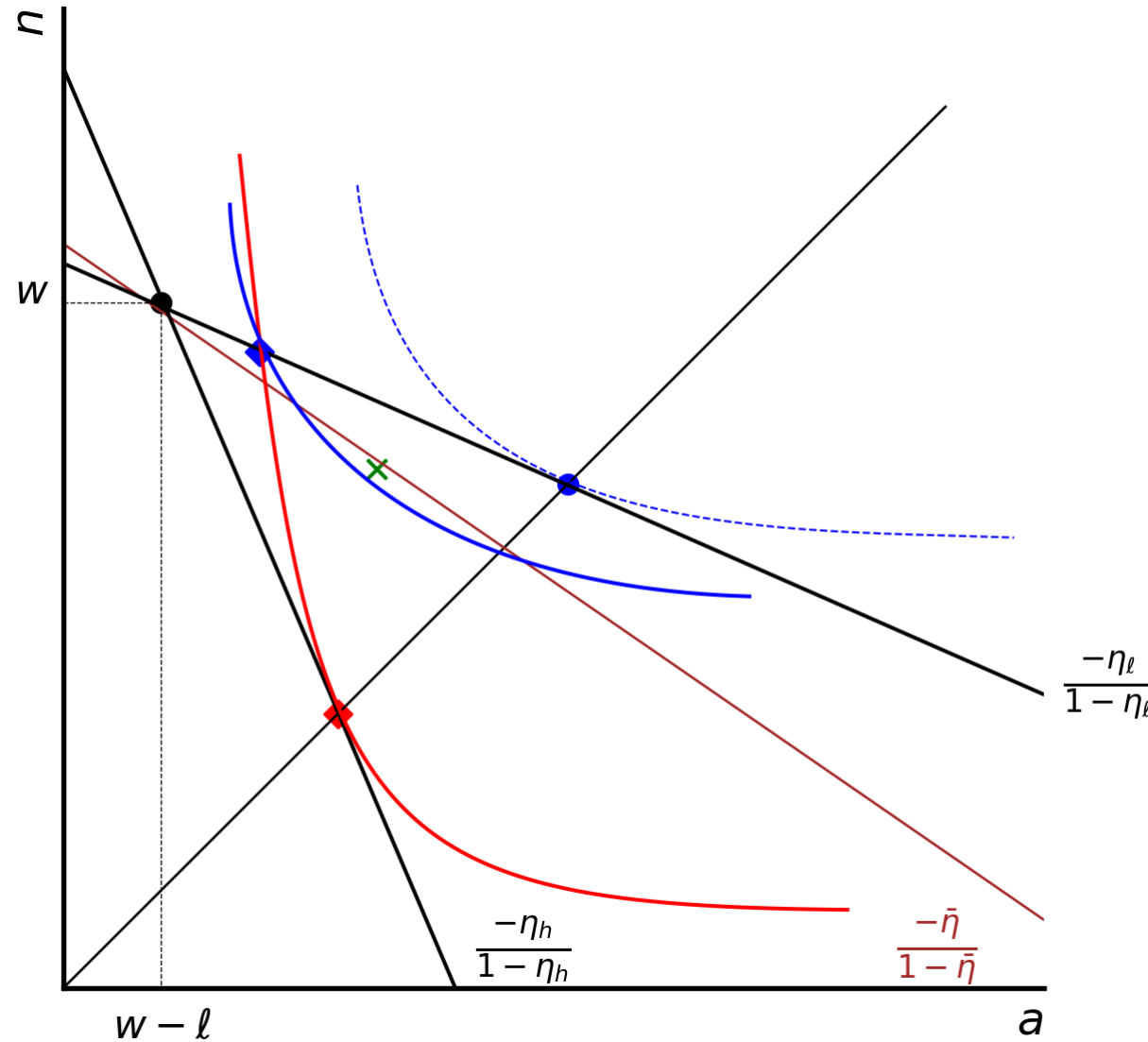
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- **Figure 1** denotes a Pareto optimal allocation—full insurance for both high-risk type and low-risk type, and zero profits for the competitive insurance industry—under full and symmetric information.
 - It is not feasible under the information structure of the example.
 - The insurance industry is not able to distinguish between high- and low-risk types. High-risk agents purchase the low-risk bundle.
 - High-risk agents purchasing the low-risk bundle generate negative profit for the insurance firm.

A separating equilibrium

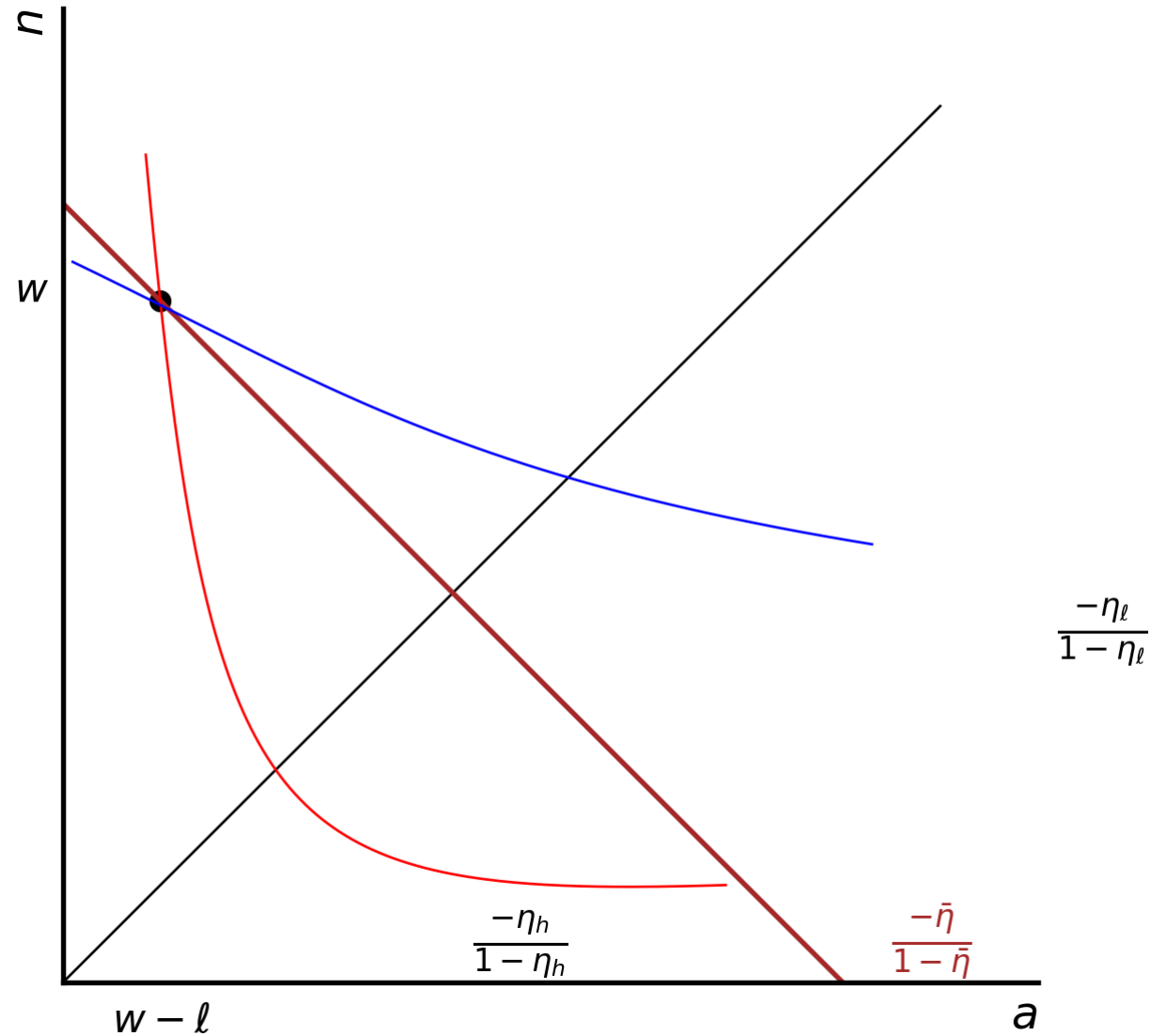


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- Figure 2 depicts a separating equilibrium.
 - High-risk agents get full insurance at high-risk price.
 - Low-risk agents are offered partial insurance at the low-risk price.
 - The insurance offered to low-risk agents is partial to prevent high-risk agent from pretending to be low-risk.

No separating equilibria

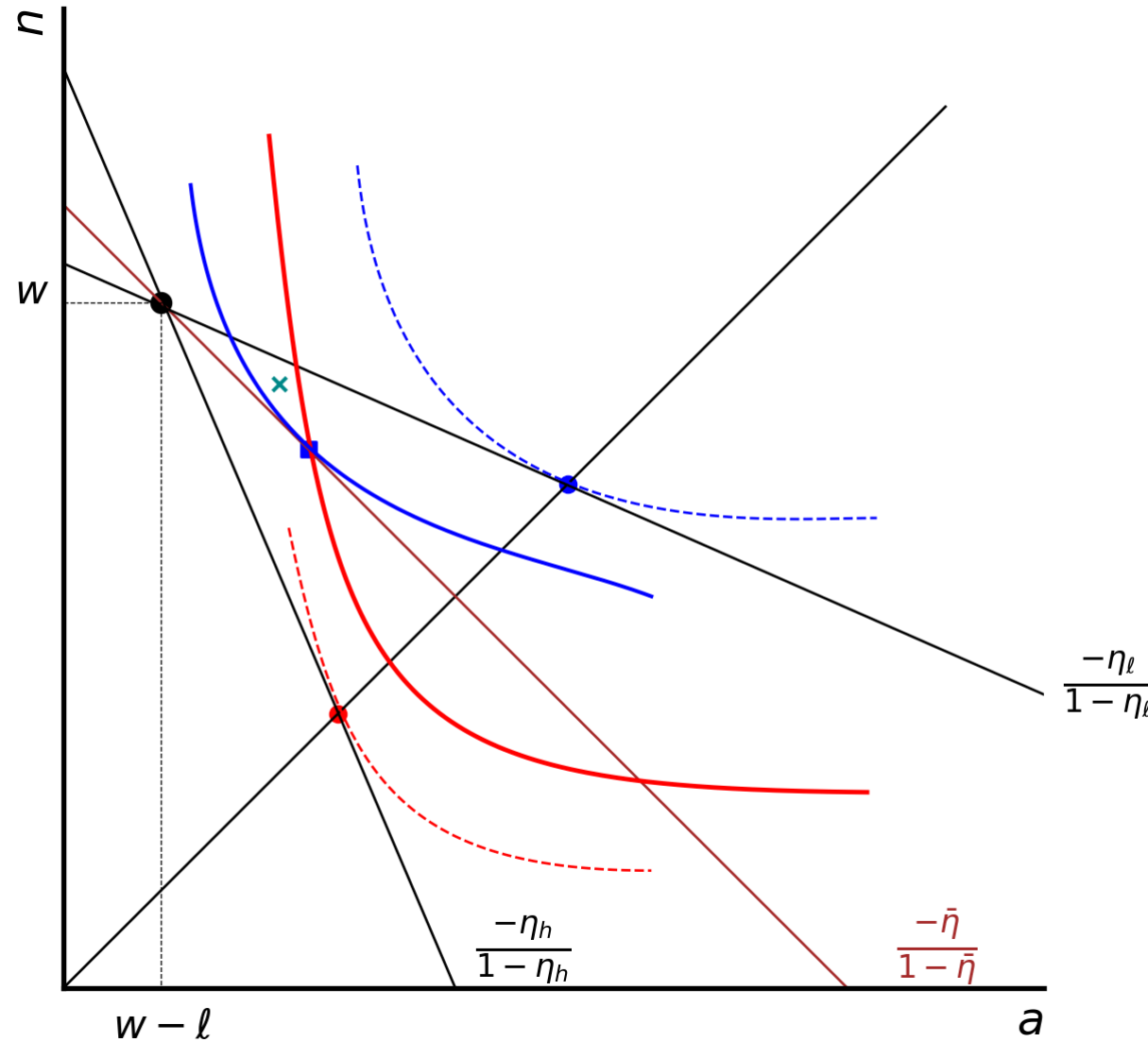


Pooling equilibria?



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- **Figure 4** depicts a situation where no pooling equilibrium exists. At pooling prices,
 - low-risk agents have no demand for insurance.
 - high-risk agents are willing to buy more than full insurance.
 - a competitive insurance market would make negative profit.
 - A possible outcome is depicted in **Figure 2**.
 - There could be full insurance for the high-risk type and partial insurance for the low-risk type at lower prices.

A pooling equilibrium with partial insurance?



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- Figure 5 appears to show a pooling equilibrium with partial insurance for both types.
 - Note however that a competitor could offer the contract depicted by x in the figure.
 - Contract x would be both profitable and attract only the low-risk agents; thus generating positive profit for the new firm.

References

Rothschild, Michael, and Joseph Stiglitz. 1976. "Equilibrium in Competitive Insurance Markets: An Essay on the Economics of Imperfect Information." *The Quarterly Journal of Economics* 90 (4): 629–49.

<http://www.jstor.org/stable/1885326>.